WIZARDOFODDS.COM

A Guide to the WC 2026 Prediction Pages

What every number, chart, and indicator means across all three pages — and how to make the most of each one.

The Foundation: One PMF, All Markets

Most football prediction tools give you a win probability and call it a day. This one gives you the **entire joint probability distribution over every possible scoreline** — the probability the match ends 0-0, that it ends 2-1, that it ends 4-3 — and derives every betting market mechanically from that single distribution.

That architecture is not a cosmetic difference. It means every number on every page is internally consistent by construction. The Over 2.5 probability and the 1X2 probabilities cannot contradict each other the way they frequently do on aggregator sites that blend data from different sources. When you see an edge on this site, you are looking at a discrepancy between a coherent probability system and the bookmaker's price — not an artifact of mismatched models.

Every page draws from the same underlying joint score PMF grid. Each cell (h, a) holds P(Home = h, Away = a): the probability the home team scores exactly h goals and the away team scores exactly a goals in regulation time. The grid sums to exactly 1.0. All markets are arithmetic operations on this grid:

Market	Formula	Interpretation
Over 2.5 goals	SUM all cells where h+a >= 3	Upper-right region of the PMF grid
Under 2.5 goals	SUM all cells where h+a <= 2	Lower-left region of the PMF grid
Both Teams to Score	SUM cells where h>=1 AND a>=1	Everything except row 0 and column 0
Home Win	SUM cells where h > a	Below the main diagonal of the grid
Draw	SUM cells where h = a	The main diagonal (0-0, 1-1, 2-2, ...)
Correct Score 2-1	P(h=2, a=1)	Single cell — read directly
P(Home scores exactly h)	SUM over all a of P(h,a)	Row marginal — collapse across all a

Page 1 — Pre-Game Predictions

sportsodds.wizardofodds.com/tools/odds-scanner/predictions/world%20cup/pre%20match.html

The command center for today's matches. Every World Cup match scheduled for today appears in the main table with the model's probability estimates, expected goals, and edge analysis against live bookmaker prices. The data refreshes automatically and a red banner appears if the underlying prediction file is more than four hours old — a signal the automated pipeline may have encountered an issue.

The KPI Cards

Four summary numbers compress the entire day's modeling output into a single glance.

Card	What It Shows	How to Read It
Matches Today	Regulation kickoffs scheduled today	Simple count. 0 = off-day or rest day.
Value Bets	Markets passing all three edge filters simultaneously	Edge >= 4% AND 90% CI lower bound > market implied AND market implied > 2%. Frequently ze
Best Edge	Single largest edge % across all markets	Specific match and market labeled below the number. Does NOT guarantee the bet will win. Confi
Avg xG / Match	Average (lambda_home + lambda_away) today	The model's expected total goals per match. 2.6 = higher-scoring slate. 2.2 = defensive slate. Use

The Bankroll Sizing Tool

Enter a bankroll amount and select a Kelly fraction. For every market passing all three edge filters, the tool computes a recommended dollar stake using the Kelly criterion: the fraction of your bankroll that maximizes the long-run growth rate of the bankroll given this edge.

Fraction	Stake Formula	Recommended When
Full Kelly	f* = Edge / (Decimal - 1)	Only when highly confident in edge accuracy. Produces large drawdowns in practice when edge esti
Half Kelly	f*/2 [default]	Standard recommendation in sports modeling literature when parameter uncertainty is non-negligible
Quarter Kelly	f*/4 [conservative]	Early in tournament when calibration sample is small, or when spreading capital across several simu

All three fractions are hard-capped at **5% of your entered bankroll** regardless of what the formula computes. This prevents extreme positions when the model identifies an unusually large apparent edge.

The Match Table

Each row represents one match. The columns, working left to right:

Column	Description
Match	Home vs. away. Neutral venue for all except USA/Canada/Mexico who receive +0.10 attack / -0.10 defense as co-hosts.
1X2 Bars	Three-segment bar: Home Win (gold), Draw (gray), Away Win (blue). Probabilities sum to 100%. Derived by summing appropriate
O/U 2.5	P(total regulation goals >= 3). Sum of all cells where h+a >= 3. Above 55% = model leans higher-scoring. Below 45% = tight matc
BTTS	P(both teams score >= 1). Sum of all cells except first row and first column. A 35% BTTS means good chance one team gets shu
Top Score	The single most probable scoreline (peak cell of the PMF) with its probability. Usually 1-0 or 1-1 carrying 12-18% in group stage.
xG (H-A)	lambda_home and lambda_away after market reconciliation. 1.8-0.7 = clear favorite-underdog. 1.2-1.1 = near coin flip.

Best Edge	Highest-edge market passing all three filters. Empty when none qualifies. Fair American odds shown are model no-margin prices.

The Expanded Row

Clicking any match row expands it to reveal three additional panels with the full detail behind the summary numbers.

Panel	Contents
Full Scoreline Distribution	All non-trivial scorelines ranked from most to least likely, with probability bars. These are raw joint PMF cell values
All Markets	Every market the engine has priced from the joint PMF: 1X2, BTTS, Over/Under at every standard line from 0.5 th
Edge Report	For each market: model probability, no-vig market implied probability (averaged across up to 6 bookmakers after S

Usage Guide

How to use Page 1: Start with the KPI cards to calibrate expectations. Scan the table for matches with a non-zero Best Edge. Expand any such match, read the full Edge Report carefully, and verify the current odds at your preferred book before placing anything. Odds change between prediction generation (8 AM UTC daily) and kickoff.

Page 2 — Probability Distributions

sportsodds.wizardofodds.com/tools/odds-scanner/predictions/world-cup/pre-match/Probability%20Distributions.html

Where Page 1 compresses model output into a single row per match, this page opens it completely. Select a match using the navigation chips at the top and every chart updates immediately to show the full probability landscape. Not just the most likely result — the entire distribution from every angle.

Chart 1 — Joint Score PMF Heatmap

The heatmap is the model in its purest form. Every cell (h, a) shows P(Home = h, Away = a). Home goals increase upward on the vertical axis; away goals increase rightward on the horizontal axis.

The color scale uses the **square root** of each cell's probability relative to the maximum cell value. This is deliberate. Without the square root, cells above ~8% would appear bright and everything else would look uniformly dark — the distribution is so heavily concentrated in the low-scoring top-left region that a linear scale is visually useless. The square root transformation separates moderate-probability cells (3–5%) from near-zero ones, making the full grid readable.

Every betting market is readable directly from this heatmap. Some examples:

What You Want to Know	Where to Look on the Heatmap
Over 2.5 goals probability	Sum of all cells in the upper-right region where h+a >= 3
BTTS probability	Everything except the first row (h=0) and first column (a=0)
Home clean sheet	First column only (a=0): all cells where away scores 0
Away clean sheet	First row only (h=0): all cells where home scores 0
Correct score 2-1	Single cell at row h=2, column a=1 — read the number
Draw probability	Diagonal cells: (0,0), (1,1), (2,2), (3,3), ...
Home win by exactly 2	Sum of cells where h-a = 2: (2,0),(3,1),(4,2),etc.

The **tail mass** shown below the heatmap is the probability assigned to scores beyond the grid boundary (more than ~8-9 goals per team). In practice vanishingly small but nonzero — it must exist so the grid sums to exactly 100%.

Chart 2 — Marginal Goal Distributions

Two bar charts side by side: the probability the home team scores exactly k goals (left), and the same for the away team (right). These marginals are derived by collapsing the joint PMF along each axis:

```
Marginal distributions:

P(Home = h) = SUM over all a of P(Home=h, Away=a) [collapse columns]
P(Away = a) = SUM over all h of P(Home=h, Away=a) [collapse rows]
```

```
# Each marginal bar chart shows P(team scores exactly k goals) for k = 0,1,2,...
# The bars must sum to 1.0 for each team independently.
```

A tall bar at k = 0 indicates this team is frequently shut out — the model assigns high probability to them failing to score at all. A relatively flat distribution across k = 1 and k = 2 reflects a strong attacking rate. In most World Cup group-stage matches, the tallest bar sits at k = 1 for both teams.

Chart 3 — Total Goals Distribution

A bar chart showing the probability of each possible total goal count from 0 through 8+. Computed by summing all cells along each anti-diagonal of the joint PMF:

```
P(Total = k) = SUM over all (h,a) where h+a=k of P(h,a)

# k=0: only cell (0,0)
# k=1: cells (1,0) and (0,1)
# k=2: cells (2,0),(1,1),(0,2)
# k=3: cells (3,0),(2,1),(1,2),(0,3) ... and so on
```

Standard over/under lines (0.5, 1.5, 2.5, 3.5, 4.5, 5.5) appear as vertical dividers on the chart. The probability to the right of any divider is the Over probability for that line; to the left is the Under. This chart makes the model's entire total-goals market visible at once — you can see Over 2.5 and Over 3.5 and Over 4.5 simultaneously without navigating to a table.

Chart 4 — Goal Difference Distribution

Centered on zero, this bar chart shows P(goal difference = d) for every possible value d = home goals - away goals. Gold bars indicate home wins (d > 0), gray indicates draws (d = 0), blue indicates away wins (d < 0).

A heavily gold-skewed distribution means the model sees the home team as a clear favorite. A roughly symmetric distribution means the match is genuinely even. The height of the gray bar relative to the gold and blue bars shows how likely a draw is given these two specific teams. High-scoring expected matches tend to have shorter gray bars — when both teams are expected to score frequently, exact ties become less probable.

Chart 5 — Top 20 Most Likely Scorelines

A ranked bar chart of the 20 most probable individual final scores with exact percentages and fair American odds shown next to each bar. The fair odds have no bookmaker margin — they represent what a perfectly calibrated market would offer for each specific score.

As a general reference: the top cell in most group-stage matches carries 12–20% probability. Any score above 6% is in the top tier of the distribution. Values below 1% are common correct-score offerings that carry long odds.

For correct-score betting: if your sportsbook is offering American odds on scoreline S, first convert to a no-vig fair probability (divide 1 by the decimal equivalent after removing the bookmaker margin), then compare to the model's percentage for that cell. A book offering +550 on a score the model gives 12% probability implies a fair price of +733 — a meaningful gap worth investigating.

O/U Lines Table

A clean table listing Over and Under probabilities for every standard total line from 0.5 through 6.5. All values are computed directly from the joint PMF — no additional modeling involved. Useful for quickly comparing the model's view across all lines without doing arithmetic from the chart.

> **Usage Guide**
> How to use Page 2: After finding an interesting match on Page 1, come here for the full picture. The heatmap shows where probability is concentrated. The goal difference chart shows how result probabilities break down. The top scorelines chart helps evaluate correct-score prices at your book. The total goals chart shows all O/U markets simultaneously.

Page 3 — Live In-Play PMF

sportsodds.wizardofodds.com/tools/odds-scanner/predictions/world-cup/live/Probability%20Distributions.html

This page activates when a World Cup match is in progress. When no match is live, it displays the next scheduled kickoff. All probabilities are regulation time only — 90 minutes plus stoppage time. Extra time and penalty shootouts are not in scope.

How the Live Model Differs from Pre-Game

The pre-game model asks: *what will the final score be from kickoff?* The live model asks something fundamentally different: **given that the current score is H-A at minute t, what will the final score be at the 90th minute?**

That distinction is fundamental. Once a match is in progress, some final scores have become impossible. If the match is 2-1 in the 70th minute, the final score cannot be 1-0, 0-0, 2-0, or any score where home is below 2 or away is below 1. Those grid cells are locked at zero probability. The remaining probability — 100% minus whatever was locked in those impossible cells — redistributes entirely across the reachable outcomes.

This is conditional probability in action. It is what separates a proper live model from a pre-game model that simply gets refreshed on a 60-second timer. The live model's grid is a fundamentally different distribution — a conditional PMF rather than a prior PMF — and the two should never be confused.

The live model also changes how it estimates remaining goal rates. Rather than assuming a constant rate across 90 minutes, it uses a **non-homogeneous hazard model**: the goal-scoring rate varies by match minute, calibrated from the minute-by-minute goal distribution across 128 World Cup matches from 2018 and 2022.

Match Phase	Minutes	Goal Rate Pattern	Explanation
Early game	1-10	Below average	Teams settling in; low pressing intensity
First half build-up	25-40	Rising	Matches open up as confidence builds
Half-time transition	45-50	Elevated	Both teams fresh; first 5 min of 2nd half active
Closing phase	75-90+	Above average	Teams chasing, committing forward; tired legs

Score-State Multipliers

On top of the temporal baseline, score-state multipliers adjust each team's goal rate based on the current scoreline. A home team losing by one goal in the 70th minute will push forward, increasing both its own attack rate and the away team's counter-attacking opportunity. These multipliers are calibrated from World Cup data and the academic football forecasting literature (Dixon and Robinson, 1998):

Score State	Home Rate	Away Rate	Primary Effect
Tied at minute 60 or later	x1.10	x1.10	Both teams press for a winner; match becomes more open

Home team losing by exactly 1	x1.25	x1.05	Home pushes forward; away positions for the counter-attack
Home team losing by 2 or more	x1.40	x1.10	Home commits to attack; significant spaces available
Home team winning by exactly 1	x0.90	x1.10	Home defends lead; away increases forward pressure
Home team winning by 2 or more	x0.80	x1.15	Home manages the game; away committed to reducing deficit

When live expected goals (xG) data is available from BallDontLie, the model blends the live xG-derived rate (60% weight) with the pre-game prior (40% weight). This blend activates at minute 15 — before that, live xG from only a handful of shots is too noisy to be useful. By minute 70, the live model is genuinely tracking the actual shot volumes and quality observed in the match, not just the pre-game forecast.

```
From minute 15 onward:

lambda_effective = 0.60 * lambda_xg_live + 0.40 * lambda_pre_game

# lambda_xg_live = xG-based rate derived from actual shots in current match
# lambda_pre_game = original pre-game expected rate from composite prior
# Blend allows the model to track actual match intensity in real time
```

Connection Badge

A small indicator shows how the page is receiving live data:

Badge	Color	Meaning
WebSocket Active	Green	Browser has an active push connection to the live prediction server. When a goal or status change is reported, t
Polling Active	Yellow	Push connection unavailable. The page fetches updated data from a static JSON file every 60 seconds. Updates

Live KPI Cards

Card	What It Shows	Normal Value
Matches Live	World Cup matches currently in progress	0-4 during match window
Next Kickoff	Next scheduled match with Eastern Time	Used when no live match
Goals Today	Total goals across live/recent matches	Accumulates through the day
Data Age	Time since last live snapshot	Under 2 min = normal; Over 10 min triggers warning

Win Probability Bar

The same three-segment bar as on Page 1 — Home Win (gold), Draw (gray), Away Win (blue) — but now **conditional on the current score and minute**. These are conditional probabilities: P(outcome | current_score = H-A, minute = t). They update with every new live snapshot.

Watch this bar during a live match and you see the model's view of the contest shifting in real time. A home goal pushes the gold segment rightward immediately. A late away equalizer can collapse the home win probability from 80% to 30% in a single update.

Pre-Game to Live Shift Table

Directly below the win probability bar, for each main market: the pre-game probability, the current live conditional probability, and the arithmetic difference between them. A large positive shift on Home Win combined with large negative shifts on Draw and Away Win tells you the home team has taken control of a match that was expected to be closer. This table is one of the most informative features on the live page — it quantifies exactly how the match is deviating from the pre-game expectation.

Win Probability Sparkline

A compact line chart showing the home team's win probability from kickoff to the current minute. This history accumulates in your browser session and resets on page reload — it is stored client-side, not on the server. Sharp upward jumps correspond to home goals; sharp downward drops correspond to away goals. The sparkline makes visible what the current snapshot cannot: whether the home team earned their current probability gradually through sustained pressure, or in a single dramatic moment, and how volatile the match has been.

Live Joint Score PMF Heatmap

The same heatmap structure as on Page 2, updated with every live snapshot and annotated with two visual cues that distinguish it from the pre-game version:

Visual Cue	Description
Red cell outline	The cell corresponding to the current live score is outlined in red. This is the 'current state' marker — the match is here
Dark unreachable cells	All cells where home goals < current home score OR away goals < current away score are forced to zero and appear d

As the match progresses, the dark region grows and the probability concentrates into fewer and fewer bright cells. In stoppage time of a 1-0 match, the heatmap may assign 85–90% probability to the single 1-0 cell, with small amounts on 1-1 (away equalizes), 2-0 (home extends), and a handful of other reachable outcomes.

Next Goal Probabilities

Three values derived from the remaining expected goals lambda_h_rem and lambda_a_rem:

```
Home scores next:
lambda_h_rem / (lambda_h_rem + lambda_a_rem)
= P(next goal is a home goal | at least one more goal scored)
```

```
Away scores next:
lambda_a_rem / (lambda_h_rem + lambda_a_rem)
= P(next goal is an away goal | at least one more goal scored)

No more goals:
e^(-lambda_h_rem) x e^(-lambda_a_rem)
= P(both remaining Poisson processes produce exactly 0 goals)
= P(current score IS the final score)

# As the match enters stoppage time with shrinking remaining lambdas,
# 'No more goals' typically climbs above 80-90%.
```

Home/Away scores next: These are conditional probabilities given that a goal will be scored — they sum to 1.0 between them and ignore the possibility of no more goals. Use them to evaluate next-scorer and first-to-score live markets.

No more goals: The joint Poisson probability that both remaining goal processes produce zero goals. This is the probability the match ends at the current score. Compare against your book's live Under line for the current total.

Top 10 Most Likely Final Scores (Live)

The same ranked scoreline list as on Page 2, restricted to reachable outcomes only. The current live score is marked with an arrow. As the match approaches the final whistle, the probability on the leading score climbs rapidly — in an 88th-minute 1-0 match, the 1-0 cell may carry 8 or 9 times the probability it held before kickoff.

This list tells you not just what is most likely but how much more likely it is than the alternatives. A top entry with 85% probability means the match is essentially decided. A top entry with 45% means there are still meaningful alternatives in play.

> **Usage Guide**
> How to use Page 3: Most valuable during fast-moving matches where the pre-game expectation is being rapidly revised — a 2-0 scoreline in the 30th minute reshapes the entire distribution. Also useful for evaluating live betting markets: compare the 'No more goals' probability against your book's live Under line; compare Next Goal fractions against live next-scorer pricing. The model's conditional probabilities update every 90 seconds during match windows.

Common Questions About the Prediction Pages

Answers to the most common questions about interpreting the numbers and using the prediction pages effectively.

Why does the model sometimes show 0 value bets?

Because the World Cup market is one of the best-covered betting markets in the world. Professional syndicates, sharp books, and quantitative models all focus heavily on these matches. The result is that the market is frequently very efficiently priced. A day with zero value bets is not a failure of the model — it is the model correctly recognizing that the bookmakers' prices are fair today.

Value bets are more likely to appear when: a match involves a team with significant recent-form information that the model has incorporated but the public has not fully priced; when a specific market (like a correct score) is listed at odds that imply a probability lower than the model's grid value; or when the model's lambda estimates differ materially from what a simple 1X2 back-calculation would suggest.

Why is the model probability for Over 2.5 different from the bookmaker's implied probability?

Several reasons. First, the model's pre-game parametric estimate (the Bivariate Poisson) and the market's price may reflect different information sets. The market may know about a key player injury that the model's lambda estimates haven't fully absorbed. Second, even after market reconciliation, the optimizer minimizes KL divergence — it moves the distribution toward the market but doesn't simply copy it.

Third, different bookmakers offer different Over 2.5 lines. The model averages across all available books after SHIN normalization; if you are comparing to a specific book's price, their margin and market position may differ from the consensus.

What does 'fair odds' mean on the site? Is that what I should expect to be offered?

Fair odds on this site are the model's no-margin American odds for each outcome — the odds a perfectly calibrated, zero-margin market would offer. In practice, every bookmaker adds a margin of 5–8% (sometimes more on exotic markets like correct scores). So the market odds available to you will typically be worse than the fair odds shown here.

The fair odds are useful for comparison: if a book is offering odds significantly better than the fair odds shown, the model sees a value opportunity. If the book's odds are significantly worse than the fair odds, the book has a large margin on that market.

The live model's probabilities changed before a goal happened — why?

The live model's conditional PMF updates based on two inputs: the current score and minute (which change the temporal hazard rate and score-state multipliers), and the live xG data (which reflects actual shot accumulation in the match). Even without a goal, the distribution shifts as time passes because the remaining expected goals shrink — a 0-0 at the 70th minute carries very different probabilities than the same 0-0 at the 25th minute.

The xG blend (60% live xG rate, 40% pre-game prior, active from minute 15) also updates continuously. If one team is dominating possession and generating high xG without scoring, the live model will reflect that elevated attacking pressure even before a goal goes in.

How should I interpret the Edge Report when Best Edge is shown but the match is hours away?

Treat it as a starting point, not a final signal. The edge report is generated at 8:00 AM UTC each day. The odds used for comparison are the best available at that time. By the time a match kicks off — potentially 8 or 10 hours later — the market may have moved. The best practice is to check the current odds at your book and recompute the edge manually using the model probabilities shown in the All Markets panel. The model probabilities themselves do not change significantly unless a major injury or lineup change is announced and the hourly update incorporates it.

The Live PMF in Real Scenarios

A walk-through of how the Live In-Play PMF changes across three representative match scenarios. All probabilities are illustrative but representative of actual model output.

Scenario A: Match is 0-0 at the 60th minute

The pre-game model might have given Home Win 45%, Draw 28%, Away Win 27%. At 0-0 in minute 60, the conditional PMF updates:

```
Remaining expected goals: lambda_h_rem ≈ 0.65, lambda_a_rem ≈ 0.50
(roughly 30 minutes remain; hazard rate increases as both teams push)
Score-state multiplier at 0-0 after min 60: both teams x1.10
lambda_h_rem_adjusted ≈ 0.72, lambda_a_rem_adjusted ≈ 0.55

Likely updated probabilities (illustrative):
Home Win: ~39% (lower than pre-game; less time for decisive goals)
Draw: ~38% (higher; 0-0 at 60 min makes draw much more likely)
Away Win: ~23% (lower for same reasons as Home Win)
No more goals: e^(-0.72) x e^(-0.55) ≈ 0.487 x 0.577 ≈ 28%
```

Scenario B: Match is 2-0 at the 30th minute

A dramatic early lead completely reshapes the distribution. The 2-0 home lead at minute 30 means scores of (0,0), (0,1), (1,0), (1,1), (2,1), (0,2) and many others are now impossible — the away team hasn't scored and the home team has 2. The remaining expected goals cover a full 60 minutes.

```
Remaining expected goals: lambda_h_rem ≈ 1.10, lambda_a_rem ≈ 0.85
Score-state multiplier: home winning by 2 -> home x0.80, away x1.15
lambda_h_rem_adj ≈ 0.88, lambda_a_rem_adj ≈ 0.98

Reachable final scores from (2,0): (2,0),(2,1),(2,2),(3,0),(3,1),(3,2),(4,0)...

Likely updated probabilities (illustrative):
```

```
Home Win: ~82% (strong favorite; large lead with 60 min remaining)
Draw: ~9% (away needs 2 goals to tie)
Away Win: ~9% (away needs 3 goals)
No more goals: e^(-0.88) x e^(-0.98) ≈ 0.415 x 0.375 ≈ 15.6%
Most likely final score: 2-0 (~30%), then 2-1 (~26%), then 3-0 (~14%)
```

Scenario C: Match is 1-1 at the 85th minute

The classic late-game tension. Most scores are now impossible; only scores reachable from 1-1 remain. The dark region of the heatmap covers most of the grid, leaving only a bright cluster near (1,1), (2,1), (1,2), and (2,2).

```
Remaining expected goals: lambda_h_rem ≈ 0.14, lambda_a_rem ≈ 0.14
Hazard rate is elevated in stoppage time, but only ~5-7 minutes remain
Score-state at 1-1 after min 60: both teams x1.10 -> ≈ 0.15 each

No more goals: e^(-0.15) x e^(-0.15) = 0.861 x 0.861 ≈ 74%

Likely updated probabilities (illustrative):
Home Win: ~11% (home needs to score; ~14% lambda, 5 min)
Draw: ~74% (most likely — no more goals)
Away Win: ~15% (away slightly more likely to score in this scenario)

Top 10 live scores: 1-1 (~74%), 2-1 (~8%), 1-2 (~12%), 2-2 (~1%), ...
```

Quick Reference: Key Symbols and Formulas

A concise reference for the mathematical notation used throughout the site and in this guide.

Symbol / Term	Definition
lambda_h	Expected home goals (rate parameter for home team's Poisson process)
lambda_a	Expected away goals (rate parameter for away team's Poisson process)
lambda_3	Shared intensity parameter in Bivariate Poisson = Cov(H,A). Currently 0.170.
lambda_h_rem	Expected remaining home goals from current minute to 90+
lambda_a_rem	Expected remaining away goals from current minute to 90+
PMF	Probability Mass Function — the full grid of P(h,a) values
T (temperature)	Calibration parameter. T=1.089 currently. T>1 flattens distribution.
SHIN normalization	Method to remove bookmaker margin from quoted odds

KL divergence	Kullback-Leibler divergence — measures how far a distribution moved
SLSQP	Sequential Least Squares Programming — constrained optimizer
NLL	Negative Log-Likelihood — primary model selection criterion
CLV	Closing Line Value — model probability vs. closing market price
Edge	(Model prob - Market prob) / Market prob. Positive = model above market.
f* (Kelly fraction)	Edge / (Decimal odds - 1). Optimal growth-maximizing bet size.
RPS	Ranked Probability Score — calibration metric for 1X2 market
ECE	Expected Calibration Error — direct probability vs. frequency measure

Comparing Model vs. Market: A Step-by-Step Example

A concrete walkthrough of how to use the Edge Report in practice. This example is illustrative but representative of real output.

Scenario: USA vs. Portugal, hypothetical group stage match.

You open Page 1 and see USA vs. Portugal listed with a Best Edge of +8.3% on 'Over 2.5 Goals'. You expand the row and read the Edge Report. Here is how to interpret each column:

Column	Value	What It Means
Market	Over 2.5	Total regulation-time goals >= 3
Model Probability	58.2%	Sum of all PMF cells where h+a >= 3, after calibration
Market Implied	53.7%	No-vig probability from 6 bookmakers, SHIN-normalized
Edge	+8.3%	(0.582 - 0.537) / 0.537 = 8.4% (proportional, not additive)
Fair Odds	-139 Am	1/0.582 - 1 expressed as American: +72 / -72... approx -139
Market Odds	-120 Am	Typical current bookmaker line for Over 2.5
CI Lower Bound	54.1%	With lambda perturbed -12%: still above market's 53.7%
Kelly Stake	2.1%	Half-Kelly at default setting: f* = 8.3%/(1.833-1)/2

This market passes all three filters: edge (8.3% >= 4%), CI lower bound (54.1% > 53.7%), and market implied (53.7% > 2%). The Kelly Stake of 2.1% means the tool recommends betting 2.1% of your bankroll at the half-Kelly fraction. At a $500 bankroll, that is $10.50.

Critical step before acting: verify the current Over 2.5 odds at your book. If the market has moved to -130 since the model ran, the new implied probability is 1/(1+100/130) ≈ 56.5%, and the edge narrows to (58.2% - 56.5%) / 56.5% = 3.0% — below the 4% threshold. The bet no longer qualifies. This is why you always check current odds.

Reading the Heatmap: What Each Region Tells You

A quick guide to what different regions of the Joint Score PMF Heatmap mean for different betting markets.

Heatmap Region	Market It Represents	What to Look For
Top-left cell (0,0)	Nil-nil draw probability	A bright cell = meaningful probability of a goalless draw
First row (h=0, all a)	Away team scores; home clean sheet	Sum of first row = P(home scores 0)
First column (a=0, all h)	Home team scores; away clean sheet	Sum of first column = P(away scores 0)
Main diagonal cells	Draw probability	Sum of (0,0),(1,1),(2,2)... = total draw %
Upper-left 3x3 corner	Matches with 0-2 total goals	Bright here = strong Under 2.5 lean
Upper-right region	Over 2.5 goals	Sum of cells where h+a >= 3 = Over 2.5 %
Top few cells of column 0	P(Away=0, Home=1,2,3...)	Home win by clean sheet probability
Single bright cell	Dominant most-likely scoreline	Correct score value — compare to book's odds

How the Three Pages Work Together

The three pages are designed to be used sequentially, from macro to micro:

Page	Purpose	Start Here When
Page 1 (Pre-Game)	Overview of today's matches. KPI cards, match table with probabilities, edge report for all markets.	You want to know which matches today have value? What are today's ke
Page 2 (Distributions)	Full probability distribution for one selected match. Heatmap, marginals, total goals, goal difference, top scorelines.	You've found a match of interest on Page 1 and want the full picture: how
Page 3 (Live)	Conditional live PMF during an in-progress match. Score-state multipliers, win probability track and sparkline, next goal prob	A match is in progress and you want to track live probabilities, evaluate

Tips for Getting the Most from the Prediction Pages

Bookmark the correct page for your workflow. If you primarily want edge alerts, bookmark Page 1. If you regularly evaluate correct-score markets, bookmark Page 2. If you watch live matches with live betting open, pre-load Page 3 before kickoff so the browser session's sparkline captures the full match arc from minute 0.

Use the marginal charts to evaluate team-level markets. The marginal distributions on Page 2 show P(home team scores exactly k goals) and P(away team scores exactly k goals) directly. These are exactly the distributions for team-total over/under markets — for example, Home Team Over 1.5 is P(home scores 0) + P(home scores 1) subtracted from 1.0.

The goal difference chart is the fastest read for result markets. On Page 2, the goal difference distribution plots show P(home wins by 1 goal), P(home wins by 2), P(draw), P(away wins by 1) etc. directly. These are the same as Asian handicap probabilities. P(difference = +1) is the probability the home team wins by exactly one goal — directly relevant for Asian handicap -1 markets.

The live page's win probability sparkline resets on reload. If you want to track the full match arc, do not reload the page mid-match. Open the live page before kickoff and let it accumulate. The sparkline data lives only in your browser session's memory.

Treat the Top 20 Scorelines chart as a correct-score value scanner. The fair American odds shown next to each bar on Page 2 have zero margin. If you can find a book offering better odds than shown for a high-probability scoreline, that is genuine value. Compare the fair odds column to what your book shows on their correct score market. Differences of more than 30 American odds points on a scoreline with 8-15% probability are worth investigating further.

The Data Age indicator on Page 3 is your health check. Under 2 minutes means the pipeline is healthy and running normally. Between 2 and 10 minutes can occur during high-demand periods or between live match windows. Above 10 minutes triggers the health warning banner and means the live page data may be stale. Do not make live betting decisions based on data that is more than 5 minutes old in a fast-moving match.

Scope and Limitations — All Pages

These limitations are real. Read them before using any number from this site to inform a betting decision.

Regulation time only. All probabilities on all three pages represent 90 minutes plus stoppage time. Extra time and penalty shootouts are not included. For knockout matches that go to extra time, the regulation-time distribution is not the complete picture.

lambda_3 = 0.170 reduces but does not eliminate the independence approximation. The Bivariate Poisson captures average correlation across all match types. Live score-state multipliers handle tactical dynamics in real time. But the pre-game prior is still a parametric approximation, not a complete model of football's complexity.

Live pipeline latency. The target of under 200 ms for WebSocket updates applies under normal conditions. Network latency and server load can affect real-time delivery. During peak periods (multiple simultaneous live matches), latency may increase.

Small calibration sample. Calibration temperature T and the WC_AVG scaling factor are estimated from limited completed 2026 match data. All parameters will stabilize as the tournament progresses to 48 and then 64 total matches.

Odds move between prediction and kickoff. Predictions are generated at 8:00 AM UTC daily. By the time you read the edge report, hours may have passed and market prices may have shifted significantly. Always verify current odds at your book before acting on any signal shown here.

Edge estimates are not profit guarantees. The model is probabilistic. Over a large sample of well-identified value bets, a model with genuine edge should show positive returns. Over any individual bet or short run, anything can happen. Never bet more than you can afford to lose.

All probabilities represent regulation time (90 minutes + stoppage time) only. Extra time and penalties are excluded. This guide is for informational and educational purposes only. Please gamble responsibly.